10d — CAP Theorem

Key Concepts

Term	Definition
CAP Theorem	A distributed system can guarantee only 2 of 3: Consistency, Availability, Partition Tolerance
Consistency (C)	Every read returns the most recent write — no stale data
Availability (A)	Every request gets a response — no timeouts or errors
Partition Tolerance (P)	System keeps working even when nodes can't communicate
Network Partition	A network failure causing nodes to lose communication
Eventual Consistency	Data will become consistent across nodes — but not instantly
Tunable Consistency	System lets you choose consistency level per query (e.g. Cassandra)

How It Works

P is non-negotiable:

- Network failures happen in all distributed systems
- You must tolerate partitions — so P is always required
- Real choice: **CP or AP**

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During a network partition:

CP system → stops responding rather than serve stale data
AP system → keeps responding but may serve stale data
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CP vs AP

	CP	AP
During partition	Refuses requests (unavailable)	Keeps responding (may be stale)
Guarantees	Consistency	Availability
Use when	Wrong answer worse than no answer	Uptime matters more than accuracy

	CP	AP
Examples	Zookeeper, HBase, MongoDB (default), etcd	Cassandra, DynamoDB, CouchDB
Real use case	Banking, payments, inventory	Shopping cart, social feed, product catalog

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Real Scenario

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Normal:
[Node A] ←──sync──→ [Node B]

During partition:
[Node A]  ✗  [Node B]  (network cut)

CP choice: Node A stops responding until sync restored ✅ consistent ❌ unavailable
AP choice: Node A keeps responding with potentially old data ❌ stale ✅ available
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PACELC (Senior-Level Extension)

CAP only covers behavior **during** a partition. PACELC adds:

- **Else (no partition):** tradeoff between **Latency** and **Consistency**
- Stricter consistency = more node coordination = higher latency
- Why Cassandra/DynamoDB offer tunable consistency — pick the tradeoff per query

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Industry Examples

- **Amazon DynamoDB** — AP; eventual consistency by default, strong consistency available at higher cost
- **Google Spanner** — CP globally; uses atomic clocks to minimize unavailability window
- **Cassandra (Discord, Netflix)** — AP; tunable consistency per query
- **Zookeeper (Kafka coordination)** — CP; must be consistent for leader election

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Interview Questions

Q: What does CAP theorem state? A: A distributed system can guarantee only 2 of: Consistency, Availability, Partition Tolerance.

Q: Why is Partition Tolerance non-negotiable? A: Network failures always happen in distributed systems — you must design for them, making the real choice between C and A.

Q: When do you choose CP over AP? A: When correctness is critical — banking, payments, inventory. A wrong answer is worse than no answer.

Q: When do you choose AP over CP? A: When uptime matters more than perfect accuracy — social feeds, shopping carts, product catalogs. Brief staleness is acceptable.

Q: What is eventual consistency? A: Data will become consistent across all nodes eventually — just not immediately. Used by AP systems like DynamoDB and Cassandra.

Q: What is PACELC? A: An extension of CAP — even without a partition, there's a tradeoff between latency and consistency. Stricter consistency requires more node coordination and increases latency.